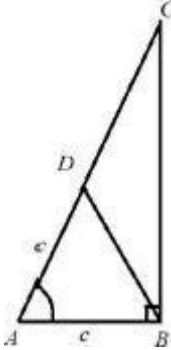
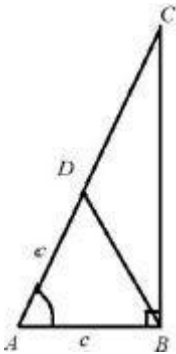


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses area of ADB = area of CDB or area of ADB = $\frac{1}{2}$ area of ABC Possibly by use of " $\frac{1}{2}ab \sin C$ " twice	AO3.1a	M1	 Area of ADB = Area of CDB $CD = AD = c$ $AC = 2c$ $\cos A = \frac{c}{2c} = \frac{1}{2}$ so $A = 60^\circ$ $\tan A = \sqrt{3}$ $\sin A = \frac{\sqrt{3}}{2}$ $2\sin A = \sqrt{3} = \tan A$
	Deduces that $AC = 2 \times AD = 2 \times AB$ or equivalent	AO1.1b	A1	
	Uses trigonometry involving sin and tan based on triangle with $AC = 2 \times AB$	AO1.1a	M1	
	Obtains correct conclusion (AG) Sets out a well-constructed mathematical argument. Use of 60° or equivalent must be justified	AO2.1	R1	
(b) (i)	Uses $\tan A = \frac{\sin A}{\cos A}$ and multiplies Or Uses sketch of two graphs to show two intersections	AO1.1a	M1	$\frac{\sin A}{\cos A} = 2 \sin A$ $\sin A = 2 \sin A \cos A$ $\sin A(1 - 2 \cos A) = 0$ $\sin A = 0$ or $\cos A = \frac{1}{2}$ $A = 0^\circ$ and $A = 60^\circ$
	Solves the equation to give $A = 0^\circ$ and 60° Or interprets intersections of graphs of the correct shape between 0° and 90° to be the solutions Special case 0° and 60° stated but not justified award B1. Stated and verified award B2	AO1.1b	A1	
(ii)	Selects $A = 60^\circ$. (Can be earned with no other	AO3.2a	B1	We need $A = 60^\circ$



working shown)			
Alternative marking Instructions	AO	Marks	Typical Solution
(a) Obtains area formula for ABD using $\sin A$	AO3.1a	M1	 Area of $ADB = \frac{1}{2}c^2 \sin A$
Obtains expression for BC using $\tan A$	AO1.1a	M1	$\frac{BC}{c} = \tan A$ $BC = c \tan A$
Obtains correct expression for area of ABC	AO1.1b	A1	Area of $ABC = \frac{1}{2}c^2 \tan A$
Simplifies to correct conclusion (AG) Sets out a well-constructed mathematical argument.	AO2.1	R1	$\frac{1}{2}c^2 \tan A = 2 \times \frac{1}{2}c^2 \sin A$ $\tan A = 2 \sin A$
(b) (i) Uses $\tan A = \frac{\sin A}{\cos A}$ and multiplies Or Uses sketch of two graphs to show two intersections	AO1.1a	M1	$\frac{\sin A}{\cos A} = 2 \sin A$ $\sin A = 2 \sin A \cos A$
Solves the equation to give $A = 0^\circ$ and 60° Or interprets intersections of graphs of the correct shape between 0° and 90° to be the solutions Special case 0° and 60° stated but not justified award B1. Stated and verified award B2	AO1.1b	A1	$\sin A(1 - 2 \cos A) = 0$ $\sin A = 0 \text{ or } \cos A = \frac{1}{2}$ $A = 0^\circ \text{ and } A = 60^\circ$
(ii) Selects $A = 60^\circ$. (Can be earned with no other working shown)	AO3.2a	B1	We need $A = 60^\circ$
Total 7 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Use $\tan \theta = \frac{\sin \theta}{\cos \theta}$ correctly (PI) (OE)	AO1.1a	M1	$\tan 15^\circ = \frac{\sqrt{2-\sqrt{3}}}{\sqrt{2+\sqrt{3}}}$
Squares and multiplies by <u>conjugate</u> appropriate <u>conjugate</u> or vice versa	AO1.1b	A1	$\begin{aligned}\tan 15^\circ &= \frac{2-\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} \\ &= (2-\sqrt{3})^2\end{aligned}$
Obtains correct values of a and b ($a = 7, b = -4$) If no explicit evidence of <u>conjugate</u> the use of <u>conjugate</u> seen award max 1/3 (M1 A0 R0)	AO2.1	R1	$\tan^2 15^\circ = 7 - 4\sqrt{3}$
			Total 3 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Begins a proof using a valid method Eg. Factor theorem, algebraic division, multiplication of correct factors	AO1.1a	M1	$p\left(-\frac{1}{2}\right) = 30 \times \left(-\frac{1}{2}\right)^3 - 7 \left(-\frac{1}{2}\right)^2 - 7 \left(-\frac{1}{2}\right) + 2$ $= 0$ $\therefore 2x + 1 \text{ is a factor of } p(x)$
	Constructs rigorous mathematical proof. To achieve this mark: Factor theorem the student must clearly substitute and state that $p(-1/2) = 0$ and clearly state that this implies that $2x + 1$ is a factor Algebraic division OR Multiplication of correct factors The method must be completely correct with a concluding statement	AO2.1	R1	
(b)	Obtains quadratic factor PI	AO1.1a	M1	$p(x) = (2x + 1)(15x^2 - 11x + 2)$



(c)	Obtains second linear factor	AO1.1b	A1	$= (2x + 1)(5x - 2)(3x - 1)$
	Writes $p(x)$ as the product of the correct three linear factors. NMS correct answer 3/3	AO1.1b	A1	
	Rearranges to achieve a cubic equation in $\sec x$ (or $\cos x$)	AO3.1a	M1	$\frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$
	Equates to zero and uses result from (b) or factorises	AO1.1a	M1	$\Rightarrow 30 \sec^2 x + 2 \cos x = 7 \sec x + 7$
	Deduces that if solutions exist they must be of the form $\sec x = -\frac{1}{2}$, $\sec x = \frac{1}{3}$ or $\sec x = \frac{2}{5}$ OE	AO2.2a	A1	$\Rightarrow 30 \sec^3 x + 2 = 7 \sec^2 x + 7 \sec x$ $30 \sec^3 x - 7 \sec^2 x - 7 \sec x + 2 = 0$
	Explains that the range of $\sec x$ is $(-\infty, -1] \cup [1, \infty)$ OE OE for $\cos x$	AO2.4	E1	$\Rightarrow (2 \sec x + 1)(5 \sec x - 2)(3 \sec x - 1) = 0$ $\Rightarrow \sec x = -\frac{1}{2}, \frac{1}{3}, \frac{2}{5}$
	Completes argument explaining that there cannot be any real solutions as values are outside of the function's range	AO2.1	R1	These values do not fall within the range of $\sec x$ as they are between -1 and 1 $\therefore \frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$ has no real solutions.
				Total 10 marks

Q4.

Marking Instructions	AO	Marks	Typical Solution
Divides or multiplies by $\cos \theta$	AO3.1a	M1	$\frac{\sin \theta \tan \theta}{\cos \theta} + 2 \frac{\sin \theta}{\cos \theta} = 3$
Obtains correct quadratic	AO1.1b	A1	$\tan^2 \theta + 2 \tan \theta - 3 = 0$
Applies a correct method to solve 'their' quadratic PI	AO1.1a	M1	$(\tan \theta + 3)(\tan \theta - 1) = 0$ $\tan \theta = 1$ or -3
Finds two correct values of $\tan \theta$ from 'their' quadratic	AO1.1b	A1F	$\theta = 45^\circ$ or 108° ALT
Obtains two correct answers CAO	AO1.1b	A1	$\sin \theta \tan \theta \cos \theta + 2 \sin \theta \cos \theta = 3 \cos^2 \theta$ $\sin^2 \theta + 2 \sin \theta \cos \theta - 3 \cos^2 \theta = 0$ $(\sin \theta + 3 \cos \theta)(\sin \theta -$



			$\cos\theta = 0$ $\tan\theta = 1 \text{ or } -3$ $\theta = 45^\circ \text{ or } 108^\circ$
Total 5 marks			

Q5.

Marking Instructions	AO	Marks	Typical Solution
Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\cot^2\theta - \cos^2\theta$
Performs some correct algebraic manipulation and uses second identity to commence proof (at least two lines of argument)	AO2.1	R1	$\equiv \frac{\cos^2\theta}{\sin^2\theta} - \cos^2\theta$ $\equiv \cos^2\theta \left(\frac{1}{\sin^2\theta} - 1 \right)$ $\equiv \cos^2\theta (\operatorname{cosec}^2\theta - 1)$ $\equiv \cos^2\theta \cot^2\theta$ $(\equiv \text{RHS})$
Concludes a rigorous mathematical argument to prove given identity AG Must start with one side and through clear logical steps arrive at the other side. In order to be sufficiently clear, each line should be a single step, unless clear further explanation is given.	AO2.1	R1	AG
			Total 3 marks

Q6.

Marking Instructions	AO	Marks	Typical Solution
(a) Makes clear attempt to use the cosine rule	AO3.1a	M1	$6^2 = 3^2 + 5^2 - 2 \times 3 \times 5 \cos\theta$
Uses trig identity with 'their' $\cos\theta$	AO1.1a	M1	$\cos\theta = \frac{3^2 + 5^2 - 6^2}{30} = -\frac{1}{15}$
Constructs rigorous argument leading to correct result AG	AO2.1	R1	$\therefore \sin\theta = \sqrt{1 - \left(-\frac{1}{15}\right)^2}$



	Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips			$\sin \theta = \frac{4\sqrt{14}}{15}$ (AG)
(b)	Writes down correct angle	AO2.2a	B1	1.64
(c)	Uses 'their' angle in $\frac{1}{2}r^2\theta$	AO1.1a	M1	$A = \frac{1}{2} \times 5^2 \times 1.64$
	Correct area FT use of incorrect obtuse angle provided both M1 marks awarded in part (a) and M1 awarded in (c)	AO1.1b	A1F	$= 20.5 \text{ m}^2$
Total 6 marks				

Q7.

(a) $x + 30^\circ = 79^\circ$, $x + 30^\circ = 180^\circ + 79^\circ$

$x = 49^\circ$

49 as the only solution in the interval $0^\circ \leq x \leq 90^\circ$

B1

$x = 229^\circ$

AWRT 229. Not given if any other soln. in the interval $90^\circ \leq x \leq 360^\circ$.

Ignore anything outside $0^\circ \leq x \leq 360^\circ$

B1

2

(b) Translation;

Accept 'translat...' as equivalent.
[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} -30^\circ \\ 0 \end{bmatrix}$$

OE Accept **full** equivalent to vector in words provided linked to 'translation / move / shift' and **correct** direction.
(0 / 2 if >1 transformation).

B1

2

(c) (i) $5 + \sin^2 \theta = (5 + 3\cos \theta) \cos \theta$

$$\Rightarrow 5 + \sin^2 \theta = 5\cos \theta + 3\cos^2 \theta$$

Correct RHS.

B1

$$5 + 1 - \cos^2 \theta = 5\cos \theta + 3\cos^2 \theta$$

$\sin^2 \theta = 1 - \cos^2 \theta$ used to get a quadratic in $\cos$.

M1

$$6 = 5\cos \theta + 4\cos^2 \theta \text{ or } 4\cos^2 \theta + 5\cos \theta - 6 (= 0)$$

ACF with like terms collected.

A1

$$\Rightarrow (4\cos \theta - 3)(\cos \theta + 2) (= 0)$$

Correct quadratic and $(4c \pm 3)(c \pm 2)$ or by formula

OE PI by $\pm$ 'correct' 2 values for $\cos \theta$.

m1

Since $\cos \theta \neq -2$, $\cos \theta = \frac{3}{4}$

CSO AG. Must show that the 'soln' $\cos \theta = -2$ has been considered and rejected

A1

5

(ii) $5 + \sin^2 2x = (5 + 3\cos 2x) \cos 2x \Rightarrow \cos 2x = \frac{3}{4}$

*Using (c)(i) to reach $\cos 2x = \frac{3}{4}$ **or** finding at least 3 solutions of $\cos \theta = \frac{3}{4}$ **and** dividing them by 2.*

M1

$$2x = 0.722(7...), 2\pi - 0.722(7...),$$

$$2\pi + 0.722(7...), 4\pi - 0.722(7...),$$

Valid method to find all four 'positions' of solutions.

m1

$$x = 0.361, 2.78, 3.50, 5.92$$

CAO Must be these four 3sf values but ignore any values outside the interval $0 < x < 2\pi$.

A1

3

[12]



Q8.

$$(a) \quad \frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \frac{\sec^2 x}{\sec^2 x - 1}$$

$\sec^2 x = 1 + \tan^2 x$ used

*M1 for correct use of $\sec^2 x = 1 + \tan^2 x$ at least once or
($\operatorname{cosec}^2 x = 1 + \cot^2 x$)*

M1

$$= \frac{\sec^2 x}{\tan^2 x} \text{ or } \frac{1 + \tan^2 x}{\tan^2 x}$$

$$\left(= \frac{1}{\cos^2 x \tan^2 x} \right)$$

$$= \frac{1}{\sin^2 x} \text{ or } \cot^2 x + 1$$

Shown, with no errors

A1

$$= \operatorname{cosec}^2 x$$

AG (No errors, omissions or poor notations seen)

A1

3

$$(b) \quad \operatorname{cosec}^2 x = \operatorname{cosec} x + 3$$

$$\operatorname{cosec}^2 x - \operatorname{cosec} x - 3 = 0$$

must have = 0

*correct solution of the quadratic,
or by completing the square*

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B1

$$\operatorname{cosec} x = \frac{1 \pm \sqrt{13}}{2} \text{ or } (2.3... \text{ and } -1.3...)$$

$$\left(\operatorname{cosec} x = \pm \sqrt{\frac{13}{4} + \frac{1}{2}} \right)$$

PI by values for $\sin x$

M1

$$\sin x = \frac{2}{1 \pm \sqrt{13}}$$

B1F for $\operatorname{cosec} x = \frac{1}{\sin^2 x}$ seen or implied

B1F



$$= 0.434 \text{ and } -0.768 \text{ (or } -0.767)$$

PI

A1

B1 for any three values correct AWRT

B1

$$x = 26^\circ, 154^\circ, -50^\circ, -130^\circ$$

*B1 for all four values correct AWRT and no extras
in the interval $-180 < x < 180^\circ$*

B1

6

(c) $2\theta - 60^\circ = x$

*where x is a written value from candidate's (b) in degrees
PI by their answer*

M1

$$\theta = 43^\circ, 5^\circ$$

CSO

A1

Ignore solutions outside interval $0^\circ < \theta < 90^\circ$

2

[11]

Q9.

(a) (i) $R = \sqrt{13}$
Accept 3.6 or better

B1

$$\tan \alpha = \frac{2}{3}$$

OE

M1

$$\alpha = 33.7^\circ$$

A1

3

(ii) minimum value = $-\sqrt{13}$
Accept -3.6 or better; ft R

B1ft

when $x - \alpha = \cos^{-1}(-1)$

NMS 0 / 2

M1

$x = 213.7^\circ$

Calculus used 0 / 2

A1

3

(b) (i) LHS = $\frac{\cos x}{\sin x} - 2 \sin x \cos x$

Express $\cot x - \sin 2x$ in terms of $\sin x$ and $\cos x$; ACF

M1

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

Factor out $\frac{\cos x}{\sin x}$ and $1 - 2 \sin^2 x$

m1

$$= \cot x \cos 2x$$

All correct

A1

3

Alternatives:

$$\text{RHS} = \cot x \cos 2x$$

$$= \frac{\cos x}{\sin x} (1 - 2 \sin^2 x)$$

Express $\cot x \cos 2x$ in terms of $\cos x$ and $\sin x$,
 $\cos 2x$ ACF

(M1)

$$= \frac{\cos x}{\sin x} - 2 \sin x \cos x$$

$\cos 2x = 1 - 2 \sin^2 x$ and multiply out and simplify.

(m1)

$$= \cot x - \sin 2x$$

All correct.

(A1)

(3)



$$\cot x(1 - \cos 2x) - \sin 2x = 0$$

$$\frac{\cos x}{\sin x} (1 - (1 - 2\sin^2 x)) - 2\sin x \cos x = 0$$

Rearrange to expression = 0 and factor out $\cot x$;

Express $\cot x$, $\cos 2x$ and $\sin 2x$ in terms of $\sin x$ and $\cos x$, ACF

(M1)

$$\frac{\cos x}{\sin x} (2\sin^2 x) - 2\sin x \cos x = 0$$

$$\cos 2x = 1 - 2\sin^2 x \text{ used}$$

(m1)

$$2\sin x \cos x - 2\sin x \cos x = 0$$

Simplified, with all correct

(A1)

(3)

(ii) $\cot x - \sin 2x = 0$

$$\cot x \cos 2x = 0$$

$$\cot x = 0 \quad \text{or} \quad \cos 2x = 0$$

Both equations correct

M1

$$2x = 90^\circ \text{ (270}^\circ\text{)}$$

Condone missing 270°

m1

$$x = 90^\circ, 45^\circ, 135^\circ$$

All correct

A1

3

Alternatives:

$$\cot x - \sin 2x = \frac{\cos x}{\sin x} - 2\sin x \cos x = 0$$

$$\cos x \left(\frac{1}{\sin x} - 2\sin x \right) = 0$$



$$\cos x = 0 \quad \text{or} \quad 1 - 2\sin^2 x = 0$$

Both equations

(M1)

$$\sin x = (\pm) \frac{1}{\sqrt{2}}$$

(m1)

$$x = 90^\circ, 45^\circ, 135^\circ$$

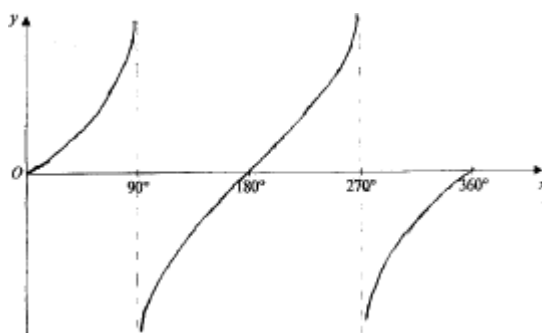
(A1)

(3)

[12]

Q10.

(a) (i)



Ignore any part of the graph drawn outside interval
 $0^\circ \leq x \leq 360^\circ$ in (a)

A 3 branch curve between 0 and 360 meeting the x-axis
 at or very close to 0, 180, 360 only

B1

A 3 branch curve between 0 and 360 with correct shape
 tending to infinity at, at least 3, of the 4 relevant ends

B1

Correct graph for $0^\circ \leq x \leq 360^\circ$ with correct intercepts.
 Asymptotes not explicitly required but graphs should
 show correct 'tendency' close to 90 and 270.

B1

3

(ii) 135° ; 315°

B2 for both 135 and 315 and no 'extra' in interval
 $0^\circ \leq x \leq 360^\circ$ (If not B2 then award B1 for either
 135 or 315 with or without extras)

B2,1,0

(c) (i) $6 \tan \theta \sin \theta = 5 \Rightarrow 6 \frac{\sin \theta}{\cos \theta} \sin \theta = 5$
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$ used

M1

$$6 \frac{\sin^2 \theta}{\cos \theta} = 5 \Rightarrow 6 \frac{1 - \cos^2 \theta}{\cos \theta} = 5$$

$\sin^2 \theta$ replaced by $1 - \cos^2 \theta$ throughout

m1

$$6 - 6\cos^2 \theta = 5\cos \theta \Rightarrow 6\cos^2 \theta + 5\cos \theta - 6 = 0$$

Completion AG Be convinced

A1

3

(ii) $6 \tan 3x \sin 3x = 5 \Rightarrow 6 \cos^2 3x + 5 \cos 3x - 6 = 0$
 Using (b)(i) with $\theta = 3x$ PI by attempting to solve
 eg for theta then dividing soln(s) by 3

M1

$$(3\cos 3x - 2)(2\cos 3x + 3) (= 0)$$

$$(\cos 3x = 2/3, -3/2)$$

Correct factorisation or correct subst into the quadratic
 formula PI by two 'correct' roots

m1

$$\cos 3x = \frac{2}{3} = \cos 48.1(89..) [= \cos \alpha]$$

$$3x = \alpha, 360 - \alpha, 360 + \alpha.$$

Dep on M1 only,

$3x = \alpha, 360 - \alpha, 360 + \alpha$ for c's α . from an eqn
 $\cos 3x = k$ where $-1 < k < 1$ OE PI and no solns
 from k outside $-1 \leq k \leq 1$

m1

$$x = 16^\circ,$$

B1

$$104^\circ,$$

B1

136°

B1

AWRT 16, 104, 136. Deduct one mark (from any award of these 3 B marks) if more than three solns given inside the interval $0^\circ \leq x \leq 180^\circ$.

Ignore any solutions outside the interval $0^\circ \leq x \leq 180^\circ$.

NMS Max. is B3/6

6

[14]

Q11.

$$(8\sec x - 2\sec^2 x = \tan^2 x - 2)$$

$$8\sec x - 2\sec^2 x = \sec^2 x - 1 - 2$$

Using $\tan^2 x = \sec^2 x - 1$ and NOT replacing $\sec^2 x$ with $1 + \tan^2 x$.

M1

$$3\sec^2 x - 8\sec x - 3 (= 0)$$

A1

$$(3\sec x + 1)(\sec x - 3) (= 0)$$

Correct factors or correct use of quadratic equation formula or completing the square for 'their' equation.

$$\sec x - \frac{8}{6} = \pm \sqrt{\frac{64}{36} + 1}$$

m1

$$\text{Or } \sec x = \frac{8 \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$$

$$\sec x = 3, -\frac{1}{3} \text{ (or } -0.33)$$

Both correct.

A1

$$\sec x = \frac{1}{\cos x}$$

PI

B1

$$\left(\cos x = \frac{1}{3} \text{ or } 0.33 \right)$$

$$\left(\sec x = -\frac{1}{3} \text{ is impossible} \right)$$

$$x = 1.23, 5.05$$

One correct. Must have earned A1 for correct quadratic, but independent of the second A1.

A1

Both correct and no extras in $0 < x < 2\pi$.

CAO

A1

[7]

Q12.

$$(a) \quad 2\sin\theta = 7\cos\theta \Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{7}{2}$$

$\frac{\sin\theta}{\cos\theta}$
 $\tan\theta = \frac{\sin\theta}{\cos\theta}$ clearly used to reach either $2\tan\theta = 7$ or
 $2/7 \tan\theta = 1$ or $\tan\theta = 3.5$ or even $\tan\theta = 2/7$ after

$$\text{seeing } \frac{\sin\theta}{\cos\theta} = \frac{2}{7}$$

M1

$$\Rightarrow \tan\theta = \frac{7}{2}$$

$$\frac{7}{2} \text{ OE eg } 3.5$$

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A1

$$(b) \quad (i) \quad 6\sin^2 x = 4 + \cos x$$

$$6(1 - \cos^2 x) = 4 + \cos x$$

$$\cos^2 x + \sin^2 x = 1 \text{ used}$$

M1

$$6 - 6\cos^2 x = 4 + \cos x$$

$$\Rightarrow 6\cos^2 x + \cos x - 2 = 0$$

CSO AG Be convinced.

A1

2

$$(ii) \quad 6\sin^2 x = 4 + \cos x \Rightarrow$$

$$6\cos^2 x + \cos x - 2 = 0$$

Uses (b)(i)

M1

$$(3\cos x + 2)(2\cos x - 1) (= 0)$$

$(3c \pm 2)(2c \pm 1)$ or by formula

M1

Correct factorisation or quadratic formula with $b^2 - 4ac$ evaluated correctly. (PI by both correct values for $\cos x$)

A1

$$\cos x = -\frac{2}{3}, \cos x = \frac{1}{2}$$

CSO Both values for $\cos x$ correct. Accept 3sf rounded or truncated.

$$x = 132^\circ, 228^\circ, 60^\circ, 300^\circ$$

B1 for any 3 of the 4 values correct. Condone greater accuracy (131.810...; 228.189...).

Ignore answers outside the given interval. Deduct 1 mark from these two B marks for each extra solution if more than 4 answers in the given interval to a min of B0
NMS: max possible is B2

B2,1,0

6

[10]

Q13.

(a) $\sec^2 \theta - 1 = \dots$

correct use of $\sec^2 \theta = 1 + \tan^2 \theta$

B1

$$\sec^2 \theta + 3\sec \theta - 10 (= 0)$$

quadratic expression in $\sec \theta$ with all terms on one side

M1

$$(\sec \theta + 5)(\sec \theta - 2) = 0$$

attempt at factors of their quadratic, $(\sec \theta \pm 5)(\sec \theta \pm 2)$, or correct use of quadratic formula

m1

$$\sec \theta = -5, 2$$

A1

$$\left(\cos \theta = -\frac{1}{5}, \frac{1}{2} \right)$$

60°, 300°, 101.5°, 258.5° (AWRT)

3 correct, ignore answers outside interval

B1

all correct, no extras in interval

B1

6

(b) $4x - 10^\circ = 60^\circ, 101.5^\circ, 258.5^\circ, 300^\circ$

$4x - 10 = \text{any of their (60)},$

M1

$4x = 70^\circ, 111.5^\circ, 268.5^\circ, 310^\circ$

all their answers from (a), BUT must have scored B1

A1F

$x = 17.5^\circ, 27.9^\circ, 67.1^\circ, 77.5^\circ$ (AWRT)

CAO, ignore answers outside interval

A1

[9]

Q14.

(a) (Area) = $\frac{1}{2} (26)(31.5) \sin \theta$

$\frac{1}{2} (26)(31.5) \sin (\theta)$ stated or used

M1

$\frac{1}{2} (26)(31.5) \times \frac{5}{13} = 157.5 \text{ (cm}^2\text{)}$

OE eg $\frac{315}{2}$

Condone AWRT 157.50

NMS: 157.5 or AWRT 157.50 scores B2

A1

2

(b) $(\cos \theta =) \frac{12}{13}$

$$\frac{12}{13} \text{ OE exact fraction}$$

B1

1

(c) $\{AC^2 =\}$
 $31.5^2 + 26^2 - 2 \times 31.5 \times 26 \times \cos(\theta) = 992.25 + 676 - 1512$
RHS of cosine rule

M1

$$= 1668.25 - 1512 = 156.25$$

Correct order of evaluation. Do not award if evaluation leads to or would lead to RHS value being outside interval 120 to 195

m1

$$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$$

12.5 OE with no sight of premature approximation clearly used

A1

3

Alternative

$$\{AC^2 =\} (26 \sin \theta)^2 + (31.5 - 26 \cos \theta)^2$$

(M1)

$$= 10^2 + 7.5^2$$

(m1)

$$AC = \sqrt{156.25} = 12.5 \text{ (cm)}$$

(A1)

(3)

[6]

Q15.

$$\tan \theta = -1$$

$$\sin^2 \theta = 3\cos^2 \theta$$

B1

$$\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta \text{ used on } \sin^2 \theta - 3\cos^2 \theta \text{ or forms and solves}$$

a correct quadratic in $\sin$ or $\cos$ and then uses to find $\tan\theta$

M1

$$\tan^2 \theta = 3$$

$$\tan^2 \theta = 3 \text{ or } \tan^2 \theta - 3 = 0$$

$$\text{or } (\tan \theta + \sqrt{3})(\tan \theta - \sqrt{3}) = 0$$

$$\text{or } \tan \theta = \sqrt{3} \text{ or } \tan \theta = -\sqrt{3}$$

A1

$$\tan \theta = \pm \sqrt{3}$$

Both

A1

[4]

Q16.

$$(a) \text{ LHS} = \frac{(1 - \cos \theta) + (1 + \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$$

Combining fractions

M1

$$= \frac{2}{1 - \cos^2 \theta}$$

Correctly simplified

A1

$$= \frac{2}{\sin^2 \theta}$$

Use of $\sin^2 \theta + \cos^2 \theta = 1$

m1

$$2 \operatorname{cosec}^2 \theta = 32$$

$$\operatorname{cosec}^2 \theta = 16$$

AG; no errors seen

A1

OR

$$1 - \cos \theta + 1 + \cos \theta = 32(1 + \cos \theta)(1 - \cos \theta) \text{ (M1)}$$

$$2 = 32(1 - \cos^2 \theta) \text{ (A1)}$$

$$2 = 32 \sin^2 \theta \text{ (m1)}$$

$$\operatorname{cosec}^2 \theta = 16 \text{ (A1)}$$

4

$$(b) \operatorname{cosec} y = (\pm) \sqrt{16} \text{ or better (PI by further working)}$$

or $\sin y = (\pm) \sqrt{\frac{1}{16}}$ or better

M1

(y =)
0.253, (2.889,) (3.394,) (6.031,) (-0.253)
Sight of any of these correct to 3dp or better

B1

(y =)
0.25, 2.89, 3.39 (or better)
Must see these 3 answers, with or without either / both of -0.25 or 6.03
Ignore answers outside interval -0.25 to 6.03 but extras in this interval scores A0

A1

x = 0.43, 1.74, 2(.00), 0.17
3 correct (must be 2 dp)
All 4 correct (must be 2 dp) and no extras in interval (ignore answers outside interval)

B1
B1

5

[9]

Q17.

(a) $\left(\frac{dx}{dy}\right) = \frac{\cos y \times \cos y - \sin y \times -\sin y}{\cos^2 y}$

Condone incorrect signs, poor notation, omission of $\frac{dy}{dx}$ or using $\frac{dx}{dy}$

M1

$$= \frac{\cos^2 y + \sin^2 y}{\cos^2 y}$$

RHS correct with terms squared, including correct notation
Must see this line

A1

$$= \frac{1}{\cos^2 y} \quad \text{or } (= 1 + \tan^2 y)$$

Must see one of these

$$\frac{dx}{dy} = \sec^2 y$$

AG; all correct including correct use of $\frac{dx}{dy}$ throughout

A1
CSO

3

(b) $\sec^2 y = 1 + (x - 1)^2$

Correct use of $\sec^2 y = 1 + \tan^2 y$ and in terms of x

M1

$$= 1 + x^2 - 2x + 1 \quad \text{OE}$$

$$= x^2 - 2x + 2$$

AG; must see " $\sec^2 y =$ ", $(x - 1)^2$ expanded and no errors seen

A1

2

(c) $\frac{dx}{dy} = x^2 + 2x + 2$ or $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

Must be seen

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

AG and no errors seen

B1

1

(d) (i) $y = \tan^{-1}(x - 1) - \ln x$

$$\left(\frac{dy}{dx} = \right) \frac{1}{x^2 - 2x + 2} - \frac{1}{x}$$

Must be correct

M1

$$\left(\frac{dy}{dx} = 0 \right)$$

$$\pm x^2 + bx + c (= 0)$$

Expression in this form (generous), where b and $c \neq 0$

m1

$$x^2 - 3x + 2 = 0$$

Must see correct equation = 0

A1

$$x = 1, 2$$

Both answers must be seen

The two A marks are independent

A1

4

(ii) $y'' = p(x^2 - 2x + 2)^{-2}(2x - 2) \pm qx^{-2}$
where p and q are constants

M1

$$y'' = -(x^2 - 2x + 2)^{-2}(2x - 2) + x^{-2}$$

p = -1, q = 1 including correct brackets

A1

2

(iii) $x = 1, y'' = 1$

Must have scored full marks in (d)(i) and (ii)

M1

At $x = 1, y'' > 0 \therefore \text{min}$

When $x = 1, y = 0$ hence on x -axis

Must see $y'' > 0$ or in words

Both statements fully correct

A1

2

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[14]

Q18.

(a) $\tan x = -3$

$$\Rightarrow = \tan^{-1}(-3) (= -71.56\dots)^\circ$$

PI eg by 71(.56..) or -71(.56..) seen

M1

$$x = 108^\circ, 288^\circ$$

Condone more accurate answers.

(108.4349..., 288.4349...).

[Ignore answers outside interval; If more than 2 answers inside interval -1 from A marks for each extra to a min of 0]

A1,A1

3

(b) (i) $7\sin^2 \theta + \sin \theta \cos \theta = 6(\cos^2 \theta + \sin^2 \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used; OE

M1

$$7\sin^2 \theta - 6\sin^2 \theta + \sin \theta \cos \theta - 6\cos^2 \theta = 0$$

$$\Rightarrow \sin^2 \theta + \sin \theta \cos \theta - 6\cos^2 \theta = 0$$

$$\Rightarrow \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\sin \theta}{\cos \theta} - 6 = 0$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta \text{ used}$$

M1

$$\Rightarrow \tan^2 \theta + \tan \theta - 6 = 0$$

CSO AG

A1

3

(ii) $(\tan \theta + 3)(\tan \theta - 2) = 0$

Factorise or other valid method to solve quadratic

M1

$$\tan \theta = -3 \text{ or } \tan \theta = 2$$

Need both

A1

$$\theta = 108^\circ, 288^\circ; \quad \theta = 63^\circ, 243^\circ;$$

Only ft on (a) for the c's two +ve $\tan^{-1}(-3)$ vals.

[B1 if 3 correct (ft)]

Condone more accurate answers.

(108.4349..., 288.4349...;

63.4349..., 243.4349...)

[Ignore answers outside interval; If more than 2 answers for each inside interval, -1 for each extra from Bs to a min of 0]

B2F,1F

4

[10]

Q19.

(a) $\cos x = -0.2$

Or $\tan x = (\pm) \sqrt{24}$

M1

$x = 1.77, 4.51$

AWRT

One correct value

A1

Second correct value and no extra values
in interval 0 to 6.28...

Ignore answers outside interval

SC

$x = 1.8, 4.5$ with or without working M1 A1 A0

SC (using degrees)

101.54, 281.54

M1 A1 A0

101.5, 281.5

M1 A0 A0

SC

No working shown

2 correct answers 3/3

1 correct answer 2/3

A1

3

(b) LHS

$$= \frac{\operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x)}{(1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x)}$$

Correctly combining fractions but
condone poor use, or omission, of brackets

M1

$$= \frac{\operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x}{1 - \operatorname{cosec}^2 x}$$

Allow recovery from incorrect brackets

A1

$$= \frac{-2 \operatorname{cosec}^2 x}{-\cot^2 x} \text{ or } \frac{-2(1 + \cot^2 x)}{-\cot^2 x}$$

Correct use of relevant trig identity

eg $\operatorname{cosec}^2 x = 1 + \cot^2 x$

m1

$$2\sec^2 x = 50$$

$$\sec^2 x = 25$$

AG

*All correct with no errors seen INCLUDING
correct brackets on 1st line*

A1

4

Or

$$\frac{\operatorname{cosec} x}{1 + \operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1 - \operatorname{cosec} x} = 50$$

$$\begin{aligned} \operatorname{cosec} x (1 - \operatorname{cosec} x) - \operatorname{cosec} x (1 + \operatorname{cosec} x) \\ = 50 (1 + \operatorname{cosec} x)(1 - \operatorname{cosec} x) \end{aligned}$$

*Correctly eliminating fractions but
condone poor use, or omission, of brackets*

(M1)

$$\begin{aligned} \operatorname{cosec} x - \operatorname{cosec}^2 x - \operatorname{cosec} x - \operatorname{cosec}^2 x \\ = 50 (1 - \operatorname{cosec}^2 x) \end{aligned}$$

Allow recovery from incorrect brackets

(A1)

$$48\operatorname{cosec}^2 x = 50$$

$$\sin^2 x = \frac{24}{25} \Rightarrow \cos^2 x = \frac{1}{25}$$

Correct use of relevant trig identity

eg $\sin^2 x = 1 - \cos^2 x$

©2025 Exam Papers Practice. All rights reserved. (m1)

$$\sec^2 x = 25$$

AG

*All correct with no errors seen INCLUDING
correct brackets on 1st line*

(A1)

(c) $\sec x = \pm 5$

Or $\cos x = \pm 0.2$

Or $\tan x = \pm \sqrt{24}$

M1

$$x = 1.77, 4.51, 1.37, 4.91$$

(AWRT)

3 correct

A1

4 correct and no other answers in interval
Ignore answers outside interval

SC

1.8, 4.5, 1.4, 4.9

With or without working

M1 A1

SC

their 2 answers from (a)

+1.37, 4.91

(AWRT)

2/3

SC For this part, if in degrees

max mark is

M1 A0

SC

No working shown

4 correct answers 3/3

3 correct answers 2/3

0, 1, 2 correct answers 0/3

A1

3

[10]

Q20.

(a) $h = 0.5$

$h = 0.5$ stated or used. (PI by x -values
0, 0.5, 1, 1.5, 2 provided no contradiction)

B1

$$f(x) = \sin x$$

$$I \approx h/2\{...\}$$

$$\{.\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of 'trapezia'..

M1

$$\{.\} =$$

$$0 + 0.90929.. + 2[0.4794.. + 0.84147.. + 0.99749..]$$

$$\{.\} = 0.90929.. + 2[2.318..] = \{ 0.90929.. + 4.636..\}$$

Min. of 2dp values rounded or truncated.
Can be implied by later correct work
provided > 1 term or a single term which
rounds to 1.39

A1

$$(I \approx) 0.25[5.546..] = 1.3865.. = 1.39 \text{ (to 3sf)}$$

CAO Must be 1.39

A1

4

- (b) Stretch(I) in y-direction(II) scale factor 2(III)

Need (I) and either (II) or (III)

M1

All correct. Need (I) and (II) and (III)

[> 1 transformation scores 0/2]

A1

2

(c) $\frac{\sin x}{\cos x} = \frac{1}{2}; \tan x = 0.5$

$$\frac{\sin x}{\cos x} = \tan x \quad \text{used to get } \tan x = k$$

or identity $\cos^2 x + \sin^2 x = 1$ used to get either $\sin^2 x = p$ or to get $\cos^2 x = q$, (p and q must be between 0 and 1)

M1

$$\tan x = 0.5$$

$$\text{Either } \tan x = \frac{1}{2} \text{ or } \cos x = \pm \sqrt{\frac{4}{5}} (= \pm 0.894..)$$

$$\text{or } \sin x = \pm \sqrt{\frac{1}{5}} (= \pm 0.447..)$$

A1

$$x = \alpha \text{ or } \pi + \alpha \text{ where } \alpha = \tan^{-1}(k)$$

Correct method to find 2nd angle. Any in wrong ft quadrants then m0. In case of squaring method candidates must also have rejected the extra 'quadrants' for the m1. Condone degrees or mixture

m1

$$x = 0.464, 3.61$$

Both. Condone > 3sf [0.463(6..), 3.60(5..or 6..)]

Accept pair of truncated values [0.463, 3.60]

Ignore any answers outside interval 0 to 6.28

A1

4

[10]

Q21.

$$\dots = 9 \sin^2 x + 6 \sin x \cos x + \cos^2 x +$$

$$\sin^2 x - 6 \sin x \cos x + 9 \cos^2 x$$

*Attempt at expanding both sets of brackets.
Minimum requirement either one of the two
expansions correct or 4 of these 6 terms seen.
Expanding and simplifying the given expression
in one step to get the correct two terms scores
this M1 and next A1*

M1

$$\dots = 10 \cos^2 x + 10 \sin^2 x$$

*Either correct pair of expansions and simplification
to remove $\sin x \cos x$ terms or full collecting of like
terms within the original correct expansion*

A1

$$= 10 (1 - \sin^2 x) + 10 \sin^2 x$$

*$\cos^2 x + \sin^2 x = 1$ clearly used. If identity is
applied correctly, but not directly, it must be
stated at the relevant point in the proof.*

M1

$$= 10 \text{ (which is an integer)}$$

*CSO [all previous 3 marks must have been
scored]*

*Condone absence of statement after 10
obtained correctly*

A1

[4]

Q22.

(a) (i) $(\sin^{-1} \pm 0.25) = \pm 14.5$

PI by sight of 194.5 etc

condone ± 14.4

M1

$$\theta = 194.5, 345.5 \text{ (AWRT)}$$

no extras in interval, ignore answers outside interval

A1

2

(ii) $2 \cot^2 (2x + 30) = 2 - 7 \operatorname{cosec} (2x + 30)$

condone replacing $2x + 30$ by Y

$$2(\operatorname{cosec}^2 (2x + 30) - 1) = 2 - 7 \operatorname{cosec} (2x + 30)$$

correct use of $\operatorname{cosec}^2 Y = 1 + \cot^2 Y$

M1

$$2 \operatorname{cosec}^2 (2x + 30) + 7 \operatorname{cosec} (2x + 30) - 4 (= 0)$$

must be in this form

A1

$$(2 \operatorname{cosec} (2x + 30) \pm 1)(\operatorname{cosec} (2x + 30) \pm 4) (= 0)$$

attempt at factorisation

m1

$$\operatorname{cosec}(2x + 30) = \frac{1}{2} \text{ or } -4$$

must be this line using $f(2x + 30)$

A1

$$2x + 30 = 194.5, 345.5$$

$$x = 82.2, 157.8 \text{ (AWRT)}$$

one correct answer, allow 82.3, ignore extra solutions

B1

CAO both answers correct and no extras in interval, ignore answers outside interval

B1

6

(b) stretch (I)

scale factor $\frac{1}{2}$ (II)

parallel to x -axis (III)

I and either II or III

M1

$$I + II + III$$

A1

translate

E1

$$\begin{pmatrix} -15 \\ 0 \end{pmatrix}$$

condone '15 to left' or '-15 in x (direction)'

B1

4

alternative method

translate

(E1)

$$\begin{pmatrix} -30 \\ 0 \end{pmatrix}$$

stretch

as above

(B1)

(M1)

scale factor $\frac{1}{2}$

parallel to x -axis

as above

(A1)

[12]

Q23.

- (a) $x + 52^\circ = (22^\circ), 180^\circ + 22^\circ; 360^\circ + 22^\circ$
 $(x = 180 + 22 - 52; x = 360 + 22 - 52)$

$x + 52 = 180 + \text{AWRT } 22, 360 + \text{AWRT } 22 \text{ OE}$

(max of M1 if extras in range)

LHS could be any letter but not x unless

final answer shows recovery

Ms can be PI

M1;M1

$$x = 150^\circ, 330^\circ$$

Both CAO with no extras in $0^\circ \leq x \leq 360^\circ$

Ignore anything outside $0^\circ \leq x \leq 360^\circ$

A1

3

(b) (i) $3 \tan \theta = \frac{8}{\sin \theta} \Rightarrow 3 \frac{\sin \theta}{\cos \theta} = \frac{8}{\sin \theta}$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ used/seen}$$

M1

$$\frac{3(1 - \cos^2 \theta)}{\cos \theta} = 8$$

$$\sin^2 \theta = 1 - \cos^2 \theta \text{ used}$$

M1

$$\Rightarrow 3 - 3 \cos^2 \theta = 8 \cos \theta$$

$$\Rightarrow 3 \cos^2 \theta + 8 \cos \theta - 3$$

CSO AG Completion

A1

3

(ii) $(3 \cos \theta - 1)(\cos \theta + 3) = 0$

Any valid method to solve the quadratic

M1

$$\cos \theta = \frac{1}{3}$$

CSO Must only be the one value

A1

2

(iii) $\cos 2x = \frac{1}{3}$

Using (ii) OE to get or use $\cos 2x = k$

where $-1 \leq k \leq 1$

M1

$$(2x =) 70.528..$$

Award for $\cos^{-1}(1/3)$ = value from 70 to

71 inclusive, even if θ used. PI

B1

$$2x = 360^\circ - 70.528.. (= 289.47...)$$

$$2x = 360 - \cos^{-1}(c's k) \text{ OE}$$

No extras inside the range

m1

$$x = 35^\circ, 145^\circ \text{ (to the nearest degree)}$$

Both, condoning greater accuracy, with

no extras in $0^\circ \leq x \leq 180^\circ$

Ignore anything outside $0^\circ \leq x \leq 180^\circ$

SC for (b)(iii) only when c's answer for

$$(b)(ii) \text{ is } \cos \theta = -\frac{1}{3} :$$

max mark M1B1 (val 70-71 or val 109-110 inclusive) m1A0

A1

4

[12]

Q24.

(a) $\sin x = \frac{1}{3}$, or sight of ± 0.34 , $\pm 0.11\pi$ or ± 19.47 (or better)

M1

$$x = 0.34, 2.8(0) \quad \text{AWRT}$$

Penalise if incorrect answers in range;
ignore answers outside range

A1

2

(b) $\operatorname{cosec}^2 x - 1 = 11 - \operatorname{cosec} x$

Correct use of $\cot^2 x = \operatorname{cosec}^2 x - 1$

M1

$$\operatorname{cosec}^2 x + \operatorname{cosec} x - 12 (= 0)$$

A1

$$(\operatorname{cosec} x + 4)(\operatorname{cosec} x - 3) (= 0)$$

Attempt at Factors

Gives $\operatorname{cosec} x$ or -12 when expanded

*Formula one error condoned*

m1

$$\left. \begin{array}{l} \operatorname{cosec} x = -4, 3 \\ \sin x = -\frac{1}{4}, \frac{1}{3} \end{array} \right\}$$

Either Line

A1

$$\sin x = -\frac{1}{4}$$

$$\Rightarrow x = 3.39, 6.03$$

AWRT

*3 correct or their two answers from (a)
and 3.39, 6.03*

B1F

$$0.34, 2.8(0)$$

AWRT

*4 correct and no extras in range
ignore answers outside range*

SC 19.47, 160.53, 194.48, 345.52 B1

B1

6

Alternative

$$\frac{\cos^2 x}{\sin^2 x} = 11 - \frac{1}{\sin x}$$

$$\cos^2 x = 11 \sin^2 x - \sin x$$

*Correct use of trig ratios and multiplying
by $2 \sin^2 x$*

(M1)

$$1 - \sin^2 x = 11 \sin^2 x - \sin x$$

$$0 = 12 \sin^2 x - \sin x - 1$$

(A1)

$$0 = (4 \sin x + 1)(3 \sin x - 1)$$

Attempt at factors as above

(m1)

$$\sin x = -\frac{1}{4}, \frac{1}{3}$$

(A1)

As above

(B1F)(B1)

[8]

Q25.

(a)
$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

M1

$$= \frac{4 \cos^2 4x + 4 \sin^2 4x}{\cos^2 4x} \text{ or better}$$

Both terms correct

A1

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

A1

or

$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

(M1)

$$= \frac{4 \cos 4x \cos 4x}{\cos 4x \cos 4x} + \frac{4 \sin 4x \sin 4x}{\cos 4x \cos 4x} \text{ or better}$$

(A1)

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

(A1)

(b) $\frac{d^2 y}{dx^2} = 4 \times 2 \tan 4x \times \dots$
 $A \tan 4x \times f(4x)$

M1

$4 \sec^2 4x$
 $f(4x) = B \sec^2 4x$

m1

$= 32 \tan 4x \sec^2 4x$
ft 8 × their p from part (a)

A1F

$= 32 \tan 4x (1 + \tan^2 4x)$
Previous two method marks must have been earned

m1

$= 32y(1 + y^2)$
 CSO

A1

Alternative Solutions

$y' = 4 + 4 \tan^2 4x = 4 + 4 \frac{\sin^2 4x}{\cos^2 4x}$
 $y'' = 4 \times \left[\frac{\cos^2 4x \cdot 2 \sin 4x \cdot 4 \cos 4x + \sin^2 4x \cdot 2 \cos 4x \cdot 4 \sin 4x}{\cos^4 4x} \right]$

$\frac{A \cos^3 4x \pm B \sin^3 4x}{\cos^4 4x}$ where A and B are constants or trig functions.

(M1)

Where A is $m \sin 4x$ and B is $n \cos 4x$

(m1)

$= \frac{4 \times 8 \sin 4x \cos 4x [\cos^2 4x + \sin^2 4x]}{\cos^4 4x}$

ft 8 × their p from part (a)

(A1F)



$$= 32 \tan 4x \sec^2 4x$$

$$k \tan 4x \sec^2 4x$$

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4 \sec^2 4x$$

$$\frac{d^2 y}{dx^2} = 4 \times 2 \sec 4x \cdot 4 \sec 4x \tan 4x$$

$$A \sec 4x \times f(4x)$$

(M1)

$$f(4x) = B \sec 4x \tan 4x$$

(m1)

$$= 32 \sec^2 4x \tan 4x$$

$$ft 8 \times \text{their } p \text{ from part (a)}$$

(A1F)

$$= 32(1 + \tan^2 4x) \tan 4x$$

Previous two method marks must have been earned

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4(1 + \tan^2 4x)$$

$$u = \tan 4x \quad \frac{dy}{dx} = 4 + 4u^2$$

$$\frac{d^2 y}{dx^2} = (8)u \frac{du}{dx}$$

(M1)

$$\frac{du}{dx} = 4 + 4 \tan^2 4x = 4 + 4u^2$$

(m1)

$$\frac{d^2y}{dx^2} = 8u(4 + 4u^2)$$

(A1)

$$= 32u(1 + u^2)$$

(m1)

$$= 32y(1 + y^2)$$

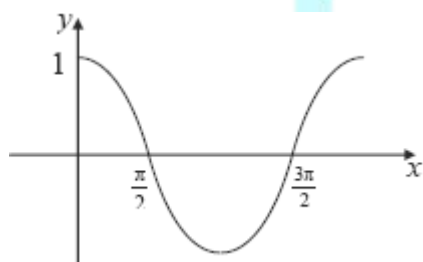
(A1)

5

[8]

Q26.

(a)



Correct shape meeting positive y-axis
and only one oscillation within interval
0 to 2π

B1

The three correct intercepts stated;
Accept 1.57 for $\pi/2$ and 4.71 for $3\pi/2$
but must be evidence of radian vals.
not just degrees

B1

Ignore any parts of the graph clearly
indicated as outside the given interval

2

(b) (i) $1 - \cos^2 \theta = \cos \theta(2 - \cos \theta)$
 $\cos^2 \theta + \sin^2 \theta = 1$ used

M1



$$1 = 2 \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$$

CSO AG Completion

A1

2

(ii) $\sin^2 2x = \cos 2x(2 - \cos 2x) \Rightarrow \cos 2x = \frac{1}{2}$

Uses (b)(i)

M1

$$\{2x =\} \cos^{-1}\left(\frac{1}{2}\right) = 1.04(7..)$$

PI Accept $1.05, \frac{\pi}{3}$; Condone 60°

m1

$$x = 0.524, 2.62$$

$$x = 0.523(59..), 2.61(7..)$$

Condone $> 3sf$; Condone $x = 0.525, 2.62$
 Accept truncated '3sf' vals $x = 0.523, 2.61$
 Deduct 1 mark for each extra (> 2 solns) in
 the given interval from A marks to a min
 of A0. Ignore any solns outside the given
 interval 0 to π .

Accept, as equivalent, the exact answers

$$x = \frac{\pi}{6} \text{ and } x = \frac{5\pi}{6} \text{ when seen and apply}$$

ISW if 'errors' converting these to decimals.

If not A2 then A1 if

- one soln correct.
- $30^\circ, 150^\circ$ ie solns left in degrees
- AWRT 0.52, 2.6 ie correct vals to only 2sf.

Must see an indication that (b)(i) has been
 used otherwise 0/4 so just stating the two
 correct answers with nothing else scores 0/4.

A2,1,0

4

[8]

Q27.

(a) $10 \operatorname{cosec}^2 x = 16 - 11 \cos x$



$$10(1 + \cot^2 x) = 16 - 11 \cot x$$

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

AG Must see evidence of correct identity
and no errors.

B1

1

- (b) Attempt at factors, giving $\pm 10 \cot^2 x \pm 6$
when expanded.

Use of formula: condone one error

M1

$$(5 \cot x - 2)(2 \cot x + 3) (= 0)$$

Correct factors

A1

$$\left(\cot x = \frac{2}{5}, -\frac{3}{2} \right)$$

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned
Condone AWRT – 0.67
ISW if x values attempted

A1,A1

Alternative 1

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

$$10 \frac{\cos^2 x}{\sin^2 x} + 11 \frac{\cos x}{\sin x} - 6 = 0$$

$$10 \cos^2 x + 11 \cos x \sin x - 6 \sin^2 x = 0$$

$$(5 \cos x - 2 \sin x)(2 \cos x + 3 \sin x) (= 0)$$

Attempt at factors, gives

$\pm 10 \cos^2 x \pm 6 \sin^2 x$ when explained

(M1)

As above

(A1)

$$(5 \cos x = 2 \sin x \quad 2 \cos x = -3 \sin x)$$

$$\frac{5}{2} = \tan x \quad -\frac{2}{3} = \tan x$$

1st A1 must be earned
Condone AWRT – 0.67
ISW if x values attempted

(A1),(A1)

Alternative 2

$$10 + 11 \tan x - 6 \tan^2 x = 0$$

$$(5 - 2 \tan x)(2 + 3 \tan x) = 0$$

Attempt at factors gives $\pm 10 \pm 6 \tan^2 x$

(M1)(A1)

$$\tan x = \frac{5}{2}, -\frac{2}{3}$$

1st A1 must be earned

Condone AWR T -0.67

ISW if x values attempted

(A1),(A1)

4

[5]

Q28.

(a) $\sec x = \frac{3}{2}$

$$\cos x = \frac{2}{3}$$

$$x = 48, 312$$

1 correct

B1

(Condone answers rounding to)

2 correct and no extras in interval

B1

2

(b) $2 \tan^2 x = 10 - 5 \sec x$
 $2 (\sec^2 x - 1) = 10 - 5 \sec x$

Use of trig identity correctly

M1

$$2 \sec^2 x + 5 \sec x - 12 (= 0)$$

A1

$$(2 \sec x - 3) (\sec x + 4) (= 0)$$

Attempt to solve or factorise

1 slip using formula

m1

$$\left. \begin{aligned} \sec x &= \frac{3}{2}, -4 \\ \cos x &= \frac{2}{3}, -\frac{1}{4} \end{aligned} \right\} \text{either of these}$$

A1

$$x = 48, 312, 104, 256$$

AWRT 3 correct condone 105 or 255

B1

All correct and no extras in interval

B1

Alternative:

$$\left. \begin{aligned} \frac{2 \sin^2 x}{\cos^2 x} &= 10 - \frac{5}{\cos x} \\ 2 \sin^2 x &= 10 \cos^2 x - 5 \cos x \\ 2 - 2 \cos^2 x &= 10 \cos^2 x - 5 \cos x \end{aligned} \right\}$$

(M1)(A1)

$$12 \cos^2 x - 5 \cos x - 2 = 0$$

then rest of scheme as above

6

[8]

Q29.

$$(a) \quad \frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\cos \theta} = 4$$

$$\tan \theta - 1 = 4$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ stated or used}$$

M1

$$\tan \theta = 5$$

AG; CSO

A1

2

$$(b) \quad (i) \quad 2 \cos^2 x - \sin x = 1$$

$$2(1 - \sin^2 x) - \sin x = 1$$

Use of $\cos^2 x + \sin^2 x = 1$

M1

$$2 - 2 \sin^2 x - \sin x = 1$$

$$\Rightarrow 2 \sin^2 x + \sin x - 1 = 0$$

AG; CSO

A1

2

(ii) $(\sin x + 1)(2 \sin x - 1) = 0$

*Factorisation or use of formula; PI by
both correct values for $\sin x$*

M1

$$\sin x = -1, \sin x = 0.5$$

Need both

A1

$$(\sin x = -1) \text{ so } x = 270^\circ$$

B1

$$(\sin x = 0.5) \text{ so } x = 30^\circ$$

30° as the only acute angle

A1

$$x = 180 - 30 = 150^\circ$$

*ft for 2nd angle from c's $\sin x = \text{non-integer}$
 Ignore values outside interval $0^\circ - 360^\circ$ but
 extras inside interval lose the corresp. B,
 A or B1F mark. If using rads, accepting
 either equivalent exact vals (in terms of
 π) or 2 dp values instead of degrees,
 penalise max of 1 mark from any of the
 final three marks (B1A1B1F) awarded
 NMS: 270° (B1); 30° , 150° (B1) [max 2/5]*

B1F

5

[9]

Q30.

(a) $\tan^{-1}\left(-\frac{1}{3}\right) = -0.32$



Sight of ± 0.32 or 18.43

M1

$$x = 2.82, 5.96$$

a correct answer AWR T

A1

–1 for any extra in range, ignore extra answers not in range.

[SC 161.57, 341.57 AWR T M1A1 (max 2/3)]

A1

3

(b) $3(\tan^2 x + 1) = 5 \tan x + 5$

$$3 \tan^2 x - 5 \tan x - 2 = 0$$

AG

B1

1

(c) $(3 \tan x + 1)(\tan x - 2) = 0$

Attempt at factorisation/formula

M1

$$\tan x = 2, -\frac{1}{3}$$

A1

$x = 1.11, 4.25, 2.82, 5.96$ AWR T

3 correct [SC $x = 1.11, 4.25$ + their two answers from (a)]

B1

4 correct, no extras in range
[SC 161.57, 341.57, 63.43, 243.43
AWR T B1 (max 3/4)]

B1

4

[8]

Q31.

(a) $\frac{3 + \sin^2 \theta}{\cos \theta - 2} = 3 \cos \theta$

$$\Rightarrow \frac{3 + (1 - \cos^2 \theta)}{\cos \theta - 2} = 3 \cos \theta$$

$\cos^2 \theta + \sin^2 \theta = 1$ stated or used [If cand starts with $\cos \theta = -\frac{1}{2}$ and gets $\sin^2 \theta = \frac{3}{4}$ without explicitly finding value for θ and verifies 1st equation is true, award M1 mo A0]

M1

$$\Rightarrow \frac{4 - \cos^2 \theta}{\cos \theta - 2} = 3 \cos \theta$$

$$\Rightarrow \frac{(2 - \cos \theta)(2 + \cos \theta)}{\cos \theta - 2} = 3 \cos \theta$$

*Difference of two squares
or division (PI by next line)*

m1

$$\Rightarrow -1(2 + \cos \theta) = 3 \cos \theta$$

A1

$$\Rightarrow -2 = 4 \cos \theta \Rightarrow \cos \theta = -\frac{1}{2}$$

CSO AG

A1

Alternative:

$$3 + 1 - \cos^2 \theta = 3 \cos^2 \theta - 6 \cos \theta$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

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$$(4 \cos \theta + 2)(\cos \theta - 2) = 0$$

Factorising or formula

(m1)

$$\cos \theta - 2 \neq 0$$

Indicates rejection of $\cos \theta = 2$

(A1)

$$\Rightarrow 4 \cos \theta = -2 \Rightarrow \cos \theta = -\frac{1}{2}$$

AG Be convinced

(A1)



(b) $\theta = 3x \Rightarrow \cos 3x = -\frac{1}{2}$

Uses part (a) to reach either $\cos 3x = -0.5$
or $\cos 3x = 0.5$

M1

$$\cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ$$

Or $\cos^{-1}(0.5) = 60^\circ$ Condone radians here

m1

$$3x = 120^\circ, 240^\circ, 480^\circ, \dots$$

$$x = 40^\circ, 80^\circ, 160^\circ$$

A1 For at least two correct.

A2, 1, 0

If > 3 solutions in the interval

$0^\circ < x < 180^\circ$, deduct 1 mark from any
A marks for each extra solution.

Deduct 1 mark from any A marks if
answers in radians. Ignore extra values
outside the given interval.

4

[8]

Q32.

(a) $\cot x = 2 \Rightarrow \tan x = 0.5$

M1

$$x = 0.46, 3.61$$

AWRT; no others within range

A1

2

(b) $\operatorname{cosec}^2 x = \frac{3 \cos x + 4}{2}$

$$2(1 + \cot^2 x) = 3 \cot x + 4$$

Correct use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$$(2 \cot^2 x - 3 \cot x + 2 - 4 = 0)$$

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

AG; correct with no slips from line

*with no fractions*

A1

2

(c) $(2 \cot x + 1)(\cot x - 2) = 0$

Attempt to solve

M1

$$\cot x = -\frac{1}{2}, 2$$

A1

$$\tan x = -2, 0.5$$

$$x = 0.46, 3.61, 2.03, 5.18$$

2 correct Allow 3.6(0)

B1

*4 correct (with no extras in range) AWRT**SC Degrees*

$$\left. \begin{array}{l} 26.57, 206.57 \\ 116.57, 296.57 \end{array} \right\} \text{B1 For 2 correct}$$

B1

4

[8]

Q33.

(a) $2x = 48$

PI by $x = 24^\circ$

B1

$$2x = 180 - 48$$

Accept equivalents for x

M1

$$2x = 360 + 48 \text{ and } 2x = 360 + 180 - 48$$

Accept equivalents for x

M1

$$x = 24^\circ, 66^\circ, 204^\circ, 246^\circ$$

CAO; need all four, no extras in given interval

A1

(b) $\frac{\sin \theta}{\cos \theta} = \tan \theta$

Stated or used

M1

$$2 \sin \theta - 3 \cos \theta = 0 \Rightarrow \tan \theta = 1.5$$

A1

$$\theta = 56.3^\circ$$

Condone > 1dp

A1

$$\theta = 56.3^\circ + 180^\circ = 236.3^\circ$$

*Ft on c's PV + 180° dep only on the M1
provided no 'extra' solutions in the given
interval.*

A1F

4

[8]

Q34.

(a) $x = \cos^{-1} \frac{1}{3}$
PI


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M1

$$= 1.23, 5.05 \text{ (} 0.39\pi, 1.61\pi \text{)}$$

*AWRT (−1 for each error in range)
SC 70.53, 289.47 B1*

A1,A1

3

(b) $\sec^2 x - 1 = 2 \sec x + 2$
use of $\sec^2 x = 1 + \tan^2 x$

M1

$$\sec^2 x - 2 \sec x - 3 = 0$$

AG; CSO

A1

2



- (c) $\sec^2 x - 2 \sec x - 3 = 0$
 $(\sec x - 3)(\sec x + 1) = 0$
attempt to solve

M1

$$\cos x = \frac{1}{3} \text{ or } -1 \quad \text{o.e.}$$

A1

$x = 1.23, 5.05,$
(2 answers in range from (a)) AWRT

B1F

3.14 (π)

all correct and no extras in range
SC 70.53, 289.47, 180 B1

B1

4

[9]

Q35.

(a) $y = \frac{\sin \theta}{\cos \theta}$

$$\frac{dy}{d\theta} = \frac{\cos \theta \cos \theta - \sin \theta (-\sin \theta)}{\cos^2 \theta}$$

$$= \frac{\pm \cos^2 \theta \pm \sin^2 \theta}{\cos^2 \theta}$$

M1A1

$$= \frac{1}{\cos^2 \theta} \quad \text{o.e.}$$

$$(1 + \tan^2 \theta)$$

$$= \sec^2 \theta$$

AG; CSO

A1

Alternative:

$$y = \frac{\tan \theta}{1}$$

$$\frac{dy}{d\theta} = \frac{1 \sec^2 \theta - 0}{1^2}$$

M1A1

$$= \sec^2 \theta$$

A1

3

(b)

$$x = \sin \theta$$

$$x^2 = \sin^2 \theta$$

$$\cos^2 \theta = 1 - x^2$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{x}{\sqrt{1 - x^2}}$$

OR LHS =

$$\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

$$= \frac{\sin \theta}{\cos \theta}$$

use of $\cos^2 \theta + x^2 = 1$ M1

$$= \tan \theta \quad \text{AG} \quad \text{AG; CSO}$$

A1

2

(c)

$$\int \frac{1}{(1 - x^2)^{\frac{3}{2}}} dx$$

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

o.e.

$$\frac{dx}{d\theta} = \pm \cos \theta$$

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M1

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$$\int = \int \frac{\cos \theta (d\theta)}{(1 - \sin^2 \theta)^{\frac{3}{2}}}$$

all in terms of θ

m1

$$= \int \frac{\cos \theta}{(\cos^2 \theta)^{\frac{3}{2}}} (d\theta)$$

A1

$$= \int \sec^2 \theta (d\theta)$$

A1

$$= \tan \theta$$

$$= \frac{x}{\sqrt{1-x^2}} (+c)$$

CSO including $d\theta$'s

A1

5

[10]



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